HAND TRACKING + ROBOTICS

WHAT IF YOUR HAND COULD MOVE A ROBOT?

A webcam turns finger movement into robot commands. No joystick needed.

Imagine moving a hot or hard-to-reach object while the robot does the reaching.



From webcam to motion in four steps

01 / SEE

Webcam

The camera watches one hand move.

02 / FIND

21 landmarks

A model estimates key points on the hand.

03 / MAP

Finger curl

Code turns the points into five curl controls.

04 / MOVE

Robot twin

A 3D hand follows the safe control values.

See the idea come alive

CyberGrip's digital hand picks up a red package, carries it across, and drops it on a platform. A successful delivery reveals the specimen.

→ GRAB / CARRY / DROP

Remote robots

People can control machines from a safer place when a task is hot, risky, or hard to reach.

Hands-free controls

Gesture controls could give people another way to interact with computers and devices.

Try it in simulation

A digital model lets builders test an idea before adding motors and hardware.

CyberGrip is a learning demo. These are possible applications, not results this project proves. The browser pickup is simulated; the optional cardboard build moves five fingers and is not medical or rescue equipment.

INSIDE THE SYSTEM

FROM PIXELS TO MOTION.

One webcam image becomes five finger controls. Here is the path the data takes.

THE LIVE HAND-TRACKING PIPELINE

01 · INPUT
Webcam
Captures a mirrored video frame. Python requests 640 × 480.

02 · VISION
MediaPipe
Finds 21 estimated landmarks on one visible hand.

03 · MATH
Finger curl
Vector angles are mapped to 0–90 curl commands.

04 · GATE
Safe check
Browser twin blocks a flagged pose or sudden wrist jerk.

05 · OUTPUT
Two paths
Three.js twin on screen, or Python serial → Arduino servos.

How does code estimate a bend?

For a finger, the program uses three landmark points. Two vectors meet at the middle joint.



$$\theta = \cos^{-1}\left(u \cdot v \,/\, |u|\,|v| \right)$$

A straighter finger is near 180°. The code maps a chosen straight-to-curled range to a control value. It is an estimate from the camera, not a real joint-angle measurement.

What the numbers mean

21 points
MediaPipe hand landmarks

5 channels
Thumb, index, middle, ring, pinky

0 → 90
Open-to-curled command values from Python

25% smoothing
Arduino eases toward each new target

The serial bridge sends packets like `<45,90,90,85,70>` at 115,200 baud.

Safety layer

In the browser demo: a raised-middle-finger pose or sudden wrist jerk turns the check red and holds the robot at its last safe pose. The hand meters still show the unfiltered estimates.

Science fair measurements

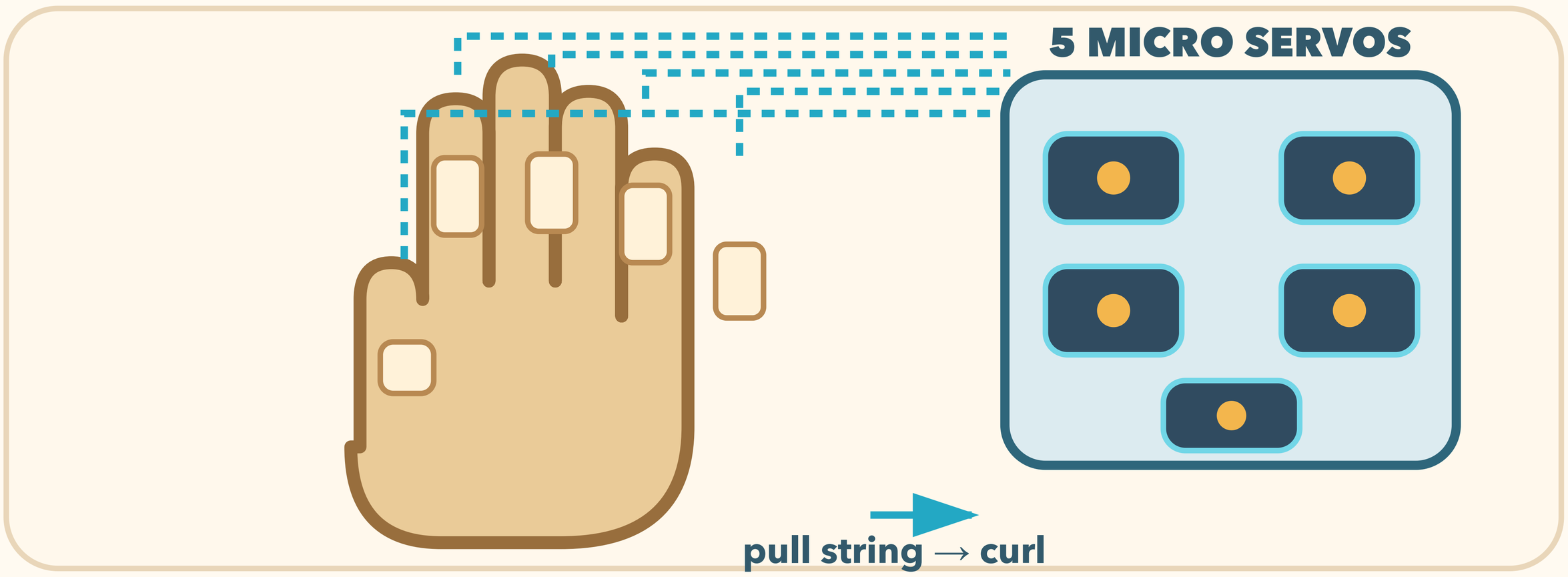
The recorder collects 100 detected-hand results with processing delay, frame rate, finger-curl estimates, camera distance, and room-light input. Change one condition at a time and compare your own trials.

Important distinction: The browser safety gate and the Python/Arduino motor path are separate. The Python motor path does not currently include that same gesture/jerk block. The web pickup is simulated; only the optional maker build moves physical servos.

MAKE THE FINGERS MOVE

BUILD A CARDBOARD ROBOT HAND.

Five small servos pull strings to copy your tracked finger curl.



Gather these parts

- ✓ Arduino Uno or Nano
- ✓ Cardboard for palm + fingers
- ✓ USB data cable
- ✓ Rubber bands or elastic
- ✓ Computer + webcam
- ✓ 5 micro servos
- ✓ 5V, 2–3A external supply
- ✓ Thread or thin string
- ✓ Tape, glue, scissors
- ✓ Jumper wires

Build in 4 steps

1 Cut the hand

Trace a palm and five cardboard fingers. Add tape hinges at each finger joint so they can bend.

2 Thread the fingers

Tape a string near each fingertip and route it behind the finger to a servo horn. Use elastic to reopen it.

3 Mount the servos

Secure five servos to a cardboard wrist/base. Start with loose strings; tighten only a little at a time.

4 Test gently

Keep the hand clear, then move one finger slowly. Adjust the string before trying the full hand.

Wire five servos

FINGER	SIGNAL WIRE
Thumb	D3
Index	D5
Middle	D6
Ring	D9
Pinky	D10

Power matters: all servo red wires → external +5V. All brown/black wires → supply GND. Connect Arduino GND to that same GND. Keep the Arduino on USB for data. **Do not power five servos from the Arduino 5V pin.**

```
Servo signal colors are often:
yellow/orange = signal · red = +5V
brown/black = GND
Check your servo label/wire guide.
```

Load the project code

1. In Arduino IDE, open `arduino/servo_controller.ino`, choose Uno/Nano + port, then upload.

2. In `python_pipeline`, install packages:

```
python3 -m pip install -r requirements.txt
```

3. Start the webcam tracker:

```
python3 tracker.py
```

4. Press `s` in the video window to connect to the Arduino. Press `q` to quit.

What the maker build does

Python estimates five finger-curl values from the webcam. The serial bridge sends them to the Arduino. The Arduino smooths the values and moves five servos.

This build copies finger motion only. It does not include the browser twin's arm or gift pickup, and the physical Python path does not have the browser demo's gesture safety filter.

Build safely with an adult

Unplug servo power before changing wires or strings. Keep fingers clear of pinches. Use the external 5V supply, check servo travel, and stop if anything gets hot or strains. Adult help for cutting and wiring.

Get the code

Type this address to open the project files:

github.com/kcmagik-blip/cybergrip-lab

Cardboard is a prototype: test one servo at a time before adding all five.